

PRASANNA SHRESTHA

Senior Software Engineer • Full-Stack & Distributed Systems

• prasannashrestha.64@gmail.com • [linkedin.com/in/shresthaprasanna](https://www.linkedin.com/in/shresthaprasanna)

SUMMARY

Senior Software Engineer with 14+ years building high-throughput distributed backend services and full-stack web applications, including 5 years at Intuit in Trust & Safety, abuse prevention, and data platforms. Led a cross-team initiative scaling a real-time fraud-risk service to 7,500 TPS, and architected spam mitigation that cut QuickBooks spam from ~40% to under 0.02% of Intuit email volume. Full-stack across Java/Spring Boot, Node.js, GraphQL, React, and TypeScript, with production GenAI and agentic-AI experience. Leads initiatives end to end, drives technical alignment across teams, and mentors engineers.

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), TypeScript, Kotlin, PHP, Python, Go, C++, SQL

Frontend: React, Redux, MobX, Angular/AngularJS, HTML, CSS, D3.js, responsive UI

Backend & Distributed Systems: Spring Boot, Node.js, Express, REST, GraphQL, microservices, Kafka, event-driven architecture, SOA, API gateway, rate limiting/throttling, caching, high availability, disaster recovery

Cloud & Infrastructure: AWS, Docker, Kubernetes, Jenkins CI/CD, Redis, Memcached, Apache Solr, Splunk

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Oracle, MS SQL Server

Testing & Quality: JUnit, Mockito, Jest, Enzyme, Playwright, Karate, WireMock; contract, integration, and E2E testing; TDD

Security & Trust/Safety: OAuth/SSO, JWT, secret rotation, fraud/abuse prevention, phishing/spam detection, LLM-based risk scoring

AI Engineering: Generative AI, agentic AI, RAG, LLM integration, MCP servers, Claude Code, AI-assisted development

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Intuit, CA Jul 2021 – Jul 2026

Trust & Safety / Product Abuse

- **Cut QuickBooks spam from ~40% to under 0.02%** of Intuit email volume by architecting multiple spam-mitigation capabilities in the real-time risk-assessment pipeline; partnered with Data Science and Product to turn abuse policies into production detection and enforcement.
- **Led the Sales Transaction Risk Assessment (STRA) service** end to end, from requirements through production, scaling it to 7,500 TPS as new clients onboarded; drove technical alignment and implementation decisions across multiple upstream and downstream teams.
- **Improved performance and API design** by consolidating 8 downstream REST APIs into a single GraphQL endpoint and adding Redis multi-tier caching, bringing per-transaction processing under 500 ms with sub-second p99 latency.
- **Built a full-stack internal rules-management tool** that lets analysts create, edit, enable, and disable wildcard-based spam rules through a real-time web UI, wired into the enforcement pipeline to block matching emails and respond quickly to new spam campaigns.
- **Shipped production GenAI and agentic-AI capabilities** — integrated LLM-based fraud detection into a pipeline processing tens of millions of outbound emails and SMS per day, using pre-check policies to reduce token usage and cost; built an agentic incident-investigation workflow that correlates service health, traffic, TPS, gateway, and Splunk signals while keeping remediation under engineer control.
- **Owned reliability for safety-critical distributed systems** — improved observability, led production incident response and RCAs, and designed overload protection, rate limiting, and disaster-recovery capabilities.
- **Raised engineering quality** by building Karate E2E, component, and WireMock contract tests and Playwright browser tests in Jenkins CI/CD; introduced Claude Code and MCP-based workflows that sped up test authoring and development.
- **Migrated a legacy PHP application** to Intuit's internal service platform (MSaaS) as sole technical owner with zero downtime despite stack incompatibilities. Mentored engineers on distributed-systems patterns and AI-assisted development. Recipient of the Engineering Excellence Award.

Tech: Java, ReactJS, TypeScript, Spring Boot, Kafka, Jenkins, Docker, AWS, DynamoDB, Kubernetes, GenAI, Agentic AI, Drools Rule Engine, CLaude Code, MCP

Software Engineer — Expedia Inc., Chicago, IL Nov 2018 – Dec 2020

- Built responsive, customer-facing UI features in React, Redux, MobX, and TypeScript in close partnership with UX, regularly shaping design decisions.
- Designed RESTful APIs and new services in Spring Boot, Node.js, and Kotlin, including third-party integrations such as a payment app in an admin tool.

- Worked with PMs and UX to refine requirements for next-generation features; maintained quality with unit tests in JUnit, Jest, and Enzyme.

Tech: React, TypeScript, Redux, MobX, Spring Boot, Node.js, Kotlin, Jest, Jenkins, Docker, AWS, PostgreSQL

Software Engineer — Cisco Systems Inc., San Jose, CA Mar 2017 – Jul 2018

- Built interactive UIs (AngularJS, React, D3.js) and Node.js server APIs with JWT authentication for a next-generation management application; contributed to system and architecture design.
- Ran CI/CD and unit testing with Jenkins and Docker, and set up VMware test environments (ESXi, vCenter, vSphere).

Tech: AngularJS, React, Node.js, D3.js, Mocha, Jenkins, Docker, OpenShift, Linux

Senior Software Developer & Team Lead — Javra Software, Nepal Jun 2015 – Apr 2016

- Led a 6-engineer team building a full-stack MEAN (MongoDB, Express, Angular, Node.js) web application across the full Agile SDLC — designed the MongoDB schema, real-time notifications (Socket.io), Redis caching, and OAuth-secured REST APIs.

Senior Software Engineer — Ace Express Travel Services, Nepal Feb 2012 – May 2015

- Designed and deployed an online flight and hotel booking engine integrating Amadeus and TBO web services; built SOAP/REST APIs, SOA-based search (JMS, Node.js), and automated CRM invoicing and reporting that reduced manual paperwork.

Web Developer — Outlines R&D; Institute of Field & Research Expedition, Nepal 2009 – 2011

- Built PHP/CodeIgniter web applications and a multimodal dictionary platform; co-authored a research paper (Oriental COCOSDA 2011).

EDUCATION

M.S. Computer Science — Maharishi University of Management, Fairfield, IA 2018

B.S. Information Management (Computer Science) — Tribhuvan University / St. Xavier's College, Nepal

CERTIFICATIONS

- AI Engineer Agentic Track: The Complete Agent & MCP Course — Udemy (Aug 2026)
- Claude Code 101 (Global) — Intuit (Mar 2026)
- Introduction to ML, Fine-Tuning, and Agentic AI — Intuit (2024–2026)
- AWS Technical Essentials — AWS (Oct 2021)

PUBLICATION

Allwood, Dhakhwa, Pokharel, Regmi, Shrestha. "Multimodal Corpus Using Multimodal Dictionary in Lohorung." Oriental COCOSDA, 2011.